

Exp No: 30

RTOS TASK SCHEDULING

```
anish@DESKTOP-BUFCRBD: ~ - X + v
GNU nano 8.7.1 rtos_scheduler.cpp *
#include <systemc.h>

SC_MODULE(rtos_scheduler) {

    void task1() {
        while (true) {
            cout << "Task1 Running at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void task2() {
        while (true) {
            cout << "Task2 Running at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(rtos_scheduler) {
        SC_THREAD(task1);
        SC_THREAD(task2);
    }
};

int sc_main(int argc, char* argv[]) {
    rtos_scheduler obj("Scheduler");
    sc_start(6, SC_SEC);
    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-J To Bracket
^X Exit      ^R Read File  ^\ Replace    ^U Paste       ^_ Justify    ^/ Go To Line M-E Redo      M-G Copy      ^B Where Was
```

```
anish@DESKTOP-BUFCRBD: ~ - X + v
anish@DESKTOP-BUFCRBD:~$ ls /usr/local/systemc
ls: cannot access '/usr/local/systemc': No such file or directory
anish@DESKTOP-BUFCRBD:~$ mkdir -p ~/exp30_rtos
cd ~/exp30_rtos
anish@DESKTOP-BUFCRBD:~/exp30_rtos$ pwd
/home/anish/exp30_rtos
anish@DESKTOP-BUFCRBD:~/exp30_rtos$ nano rtos_scheduler.cpp
anish@DESKTOP-BUFCRBD:~/exp30_rtos$ cd ~/exp30_rtos
anish@DESKTOP-BUFCRBD:~/exp30_rtos$ g++ -std=c++17 rtos_scheduler.cpp -o rtos_scheduler -I/usr/local/systemc/include -L/usr/local/systemc/lib-linux64 -lsystemc -Wl,-rpath,/usr/local/systemc/lib-linux64
anish@DESKTOP-BUFCRBD:~/exp30_rtos$ ls
rtos_scheduler  rtos_scheduler.cpp
anish@DESKTOP-BUFCRBD:~/exp30_rtos$ ./rtos_scheduler

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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ALL RIGHTS RESERVED
Task1 Running at 0 s
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
anish@DESKTOP-BUFCRBD:~/exp30_rtos$ |
```

